

◆解答◆

- | | |
|-----------------------------------|---------------------------|
| ① (1) 16.45 | (2) $\frac{2}{9}$ |
| (3) 6cm | (4) 2800 |
| (5) 7.4 | |
| ② (1) ① 50.24cm | ② 200.96cm ² |
| (2) ① 6.28cm | ② 15.7cm ² |
| (3) 113.04cm ² | |
| ③ (1) 26.84cm | (2) 102.8cm |
| ④ (1) 18.84cm ² | (2) 119.32cm ² |

◆解説◆

- ②** (1) ① $8 \times 2 \times 3.14 = 50.24$ (cm)
 ② $8 \times 8 \times 3.14 = 200.96$ (cm²)
 (2) ① $5 \times 2 \times 3.14 \times \frac{72}{360} = 6.28$ (cm)
 ② $5 \times 5 \times 3.14 \times \frac{72}{360} = 15.7$ (cm²)
 (3) 2つの円の半径は,
 $20 \div 2 = 10$ (cm)
 $(20 - 4) \div 2 = 8$ (cm)
 $10 \times 10 \times 3.14 - 8 \times 8 \times 3.14$
 $= (100 - 64) \times 3.14$
 $= 36 \times 3.14 = 113.04$ (cm²)
- ③** (1) $8 \times 3.14 \times \frac{180}{360} + 8 \times 2 \times 3.14 \times \frac{45}{360}$
 $= 18.84$ (cm) … 曲線部分
 $18.84 + 8 = 26.84$ (cm)
 (2) $10 \times 4 + 10 \times 3.14 \times 2 = 102.8$ (cm)
- ④** (1) 3つの半円の半径は,
 $(6 + 4) \div 2 = 5$ (cm)
 $6 \div 2 = 3$ (cm)
 $4 \div 2 = 2$ (cm)
 $5 \times 5 \times 3.14 \times \frac{180}{360}$
 $- 3 \times 3 \times 3.14 \times \frac{180}{360} - 2 \times 2 \times 3.14 \times \frac{180}{360}$
 $= (25 - 9 - 4) \times \frac{1}{2} \times 3.14$
 $= 6 \times 3.14 = 18.84$ (cm²)
 (2) 2つの半円の半径は,
 12 cm, $12 - 4 = 8$ (cm)
 $12 \times 12 \times 3.14 \times \frac{135}{360} - 8 \times 8 \times 3.14$
 $\times \frac{135}{360} + 8 \times 8 \times 3.14 \times \frac{180 - 135}{360}$
 $= (54 - 24 + 8) \times 3.14$
 $= 38 \times 3.14 = 119.32$ (cm²)